**Duration: You will need at least two hours for this exam. I recommend planning for four hours.**

DO NOT OPEN the exam until you are ready to work. Include COMPLETE and DETAILED JUSTIFICATIONS for all the problems.

**This exam is due no later than the start of class on Monday, May 2nd, 2022**

**You may not consult in any form with any other person while doing this take-home exam.** Do not consult any online sources. You may, however, consult our textbook, your class notes, your own homework, and resources I have posted or given out in class. **You are welcome to email me if you need clarifications on any of the problems.**

**After you have completed the exam, you should do the following:**

1. Recopy your final solutions neatly, *starting each problem on a new page.*

2. Print your name at the top of this page, and read and sign the certification below.

3. Include this cover sheet on top of your rewritten solutions. (Your solutions will not be graded without your signed cover sheet.)

**CERTIFICATION**:

Please read and sign (certify that you agree to and followed) the following statement before submitting your solutions to this exam.

**I hereby certify that this exam represents my work, and my work only, and that I did not consult with any other person in answering the questions on this exam. I have consulted only with our textbook, my class notes, and my own homework. I have not consulted any books other than our textbook.**

Signature:

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| Problem | Score |
|---|---|
| 1 *(7 pts)* | |
| 2 *(10 pts)* | |
| 3 *(5 pts)* | |
| 4 *(5 pts)* | |
| 5 *(5 pts)* | |
| 6 *(8 pts)* | |
| 7 *(10 pts)* | |
| TOTAL*(50 pts)* | |

1. Solve the 2D Heat equation on a rectangle: $u_t = 9\nabla^2 u$, on $0 \leq x \leq 1$, and $0 \leq y \leq \pi$, with boundary conditions $u(0, y, t) = u(1, y, t) = u_y(x, 0, t) = u_y(x, \pi, t) = 0$, and initial condition $u(x, y, 0) = xy^2$

2. Solve the Laplace equation $u_{xx} + u_{yy} = 0$ for $u(x, 0) = 0$, $u(x, \pi) = 1$, $u_x(0, y) = 0$, and $u_x(\pi, y) + u(\pi, y) = 1$.

3. Solve the Poisson equation $u_{xx} + u_{yy} = y\cos(x)$ for $u(x, 0) = 0$, $u(x, \pi) = 0$, $u_x(0, y) = 0$, and $u_x(\pi, y) = 0$.

4. Solve the Sturm-Liouville problem for all eigenvalues and eigenfunctions: $y'' + \lambda y = 0$, with periodic BCs $y(0) = y(L)$, and $y'(0) = y'(L)$.

5. What is the Fourier Integral Representation and how does it define the Fourier Tranform. Explain in depth.

6. Apply Fourier Transforms to the following functions: (Use the table, but show all work and justify selections)

   (a) $f(x) = \dfrac{2}{4 + x^2}$

   (b) $f(x) = \dfrac{d^n}{dx^n} g(x)$

7. Determine the solution to the given PDE. You only need to write your answer in the form of the inverse transform (you do not need to evaluate its integral.) Assume $0 < x < \infty$ and $t > 0$. $u_t = 2u_{xx}$, $u(x, 0) = e^{-2x^2}$, and $u_x(0, t) = 0$.

| | $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\alpha) e^{i\alpha x}\, d\alpha$ | | $F(\alpha) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\alpha x}\, dx$ |
|---|---|---|---|
| 1. | $f'(x)$ | 1. | $i\alpha F(\alpha)$ |
| 2. | $xf(x)$ | 2. | $iF'(\alpha)$ |
| 3. | $f(x-c)$ | 3. | $e^{-ic\alpha}F(\alpha)$ |
| 4. | $e^{icx}f(x)$ | 4. | $F(\alpha - c)$ |
| 5. | $f(cx)$ | 5. | $\frac{1}{c}F\left(\frac{\alpha}{c}\right)$ |
| 6. | $\frac{1}{c}f\left(\frac{x}{c}\right)$ | 6. | $F(c\alpha)$ |
| 7. | $f(x) * g(x)$ | 7. | $\sqrt{2\pi}\ F(\alpha)G(\alpha)$ |
| 8. | $f(x)g(x)$ | 8. | $\frac{1}{\sqrt{2\pi}}\ F(-\alpha) * G(-\alpha)$ |
| 9. | $e^{-c\lvert x\rvert}$ | 9. | $\sqrt{\frac{2}{\pi}}\ \frac{c}{\alpha^2+c^2}$ |
| 10. | $\frac{1}{x^2+c^2}$ | 10. | $\sqrt{\frac{\pi}{2}}\ e^{-c\lvert\alpha\rvert}/c$ |
| 11. | $\exp[-(x-m)^2/2\sigma^2]/\sigma\sqrt{2\pi}$ | 11. | $\exp\left[-\frac{\sigma^2\alpha^2}{2} - im\alpha\right] \Big/ \sqrt{2\pi}$ |
| 12. | $\begin{cases} 1, & \text{if } \lvert x\rvert \le L, \\ 0, & \text{if } \lvert x\rvert > L. \end{cases}$ | 12. | $\sqrt{\frac{2}{\pi}}\ \frac{\sin L\alpha}{\alpha}$ |
| 13. | $\frac{\sin Lx}{x}$ | 13. | $\begin{cases} \sqrt{\frac{\pi}{2}}, & \text{if } \lvert\alpha\rvert \le L, \\ 0, & \text{if } \lvert\alpha\rvert > L. \end{cases}$ |
| 14. | $\delta(x)$ | 14. | $\frac{1}{\sqrt{2\pi}}$ |
| 15. | $1$ | 15. | $\sqrt{2\pi}\ \delta(\alpha)$ |
| 16. | $H(x)$ | 16. | $\frac{1}{i\alpha\sqrt{2\pi}} + \sqrt{\frac{\pi}{2}}\ \delta(\alpha)$ |
| 17. | $\text{sgn } x$ | 17. | $\sqrt{\frac{2}{\pi}}\ \frac{1}{i\alpha}$ |
| 18. | $\frac{1}{x}$ | 18. | $\sqrt{\frac{\pi}{2}}\frac{\text{sgn } \alpha}{i}$ |
| 19. | $\text{sgn } x * f(x)$ | 19. | $\dfrac{2F(\alpha)}{i\alpha}$ |
| 20. | $\frac{f(x)}{x}$ | 20. | $-\frac{i}{2}\ \text{sgn } \alpha * f(-\alpha)$ |

A68 Appendix B

**Table of Fourier Cosine Transforms**

| $f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \mathcal{F}_c(f)(\omega) \cos \omega x \, d\omega, \quad 0 < x < \infty$ | $\mathcal{F}_c(f)(\omega) = \widehat{f}_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos \omega x \, dx, \quad 0 \le \omega < \infty$ |
|---|---|
| **1.** $\begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$ | $\sqrt{\frac{2}{\pi}} \frac{\sin a\omega}{\omega}$ |
| **2.** $e^{-ax}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \frac{a}{a^2+\omega^2}$ |
| **3.** $x\, e^{-ax}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \frac{a^2-\omega^2}{(a^2+\omega^2)^2}$ |
| **4.** $e^{-a\,x^2/2}, \quad a > 0$ | $\frac{1}{\sqrt{a}} e^{-\omega^2/2a}$ |
| **5.** $\cos ax\, e^{-ax}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \frac{a\omega^2+2a^3}{4a^4+\omega^4}$ |
| **6.** $\sin ax\, e^{-ax}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \frac{2a^3-a\omega^2}{4a^4+\omega^4}$ |
| **7.** $\frac{a}{a^2+x^2}, \quad a > 0$ | $\sqrt{\frac{\pi}{2}} e^{-a\,\omega}$ |
| **8.** $x^p, \quad 0 < p < 1$ | $\sqrt{\frac{2}{\pi}} \frac{\Gamma(p) \cos(p\omega/2)}{\omega^p}$ |
| **9.** $\begin{cases} \cos x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$ | $\frac{1}{\sqrt{2\pi}} \left[ \frac{\sin a(1-\omega)}{1-\omega} + \frac{\sin a(1+\omega)}{1+\omega} \right]$ |
| **Operational Properties** | |
| **10.** $\alpha f(x) + \beta g(x)$ | $\alpha \mathcal{F}_c(f)(\omega) + \beta \mathcal{F}_c(g)(\omega)$ |
| **11.** $f(ax), \quad a > 0$ | $\frac{1}{a} \widehat{f}_c(\frac{\omega}{a})$ |
| **12.** $f'(x)$ | $\omega \widehat{f}_s(\omega) - \sqrt{\frac{2}{\pi}} f(0)$ |
| **13.** $f''(x)$ | $-\omega^2 \widehat{f}_c(\omega) - \sqrt{\frac{2}{\pi}} f'(0)$ |
| **14.** $x f(x)$ | $\left[\widehat{f}_s\right]'(\omega)$ |
| **15.** $\mathcal{F}_c(\mathcal{F}_c f)$ | $f$ |

## Table of Fourier Sine Transforms

| $f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \mathcal{F}_s(f)(\omega) \sin \omega x \, d\omega, \quad 0 < x < \infty$ | $\mathcal{F}_s(f)(\omega) = \widehat{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin \omega x \, dx, \quad 0 < \omega < \infty$ |
|---|---|
| 1. $\begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$ | $\sqrt{\frac{2}{\pi}} \frac{1-\cos a\omega}{\omega}$ |
| 2. $e^{-ax}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \frac{\omega}{a^2+\omega^2}$ |
| 3. $x e^{-ax}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \frac{2a\omega}{(a^2+\omega^2)^2}$ |
| 4. $\frac{e^{-ax}}{x}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \tan^{-1} \frac{\omega}{a}$ |
| 5. $\frac{1}{2} x e^{-a x^2}, \quad a > 0$ | $\frac{\omega}{a^{3/2}} e^{-\omega^2/2a}$ |
| 6. $\cos ax \, e^{-ax}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \frac{\omega^3}{4a^4+\omega^4}$ |
| 7. $\sin ax \, e^{-ax}, \quad a > 0$ | $\sqrt{\frac{2}{\pi}} \frac{2a^2\omega}{4a^4+\omega^4}$ |
| 8. $\frac{x}{a^2+x^2}, \quad a > 0$ | $\sqrt{\frac{\pi}{2}} e^{-a\omega}$ |
| 9. $x^{p-1}, \quad 0 < p < 1$ | $\sqrt{\frac{2}{\pi}} \frac{\Gamma(p) \cos(\pi p/2)}{\omega^p}$ |
| 10. $\begin{cases} \sin x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$ | $\frac{1}{\sqrt{2\pi}} \left[ \frac{\sin a(1-\omega)}{1-\omega} - \frac{\sin a(1+\omega)}{1+\omega} \right]$ |

## Operational Properties

| | |
|---|---|
| 11. $\alpha f(x) + \beta g(x)$ | $\alpha \mathcal{F}_s(f)(\omega) + \beta \mathcal{F}_s(g)(\omega)$ |
| 12. $f(ax), \quad a > 0$ | $\frac{1}{a} \widehat{f}_s(\frac{\omega}{a})$ |
| 13. $f'(x)$ | $-\omega \widehat{f}_c(\omega)$ |
| 14. $f''(x)$ | $-\omega^2 \widehat{f}_s(\omega) + \sqrt{\frac{2}{\pi}} \omega f(0)$ |
| 15. $x f(x)$ | $-\left[\widehat{f}_c\right]'(\omega)$ |
| 16. $\mathcal{F}_s(\mathcal{F}_s f)$ | $f$ |